Introduction to Object-Oriented Analysis

Topic: Explain Class Diagrams and Object-Oriented Principles.

Introduction
Object-Oriented Analysis (OOA) focuses on identifying objects, classes, and their relationships before software design.

Class Diagram
A Class Diagram is a UML diagram representing classes, attributes, methods, and relationships such as association, inheritance, aggregation, composition, and dependency.

Main Components
Class: Blueprint for objects.
Attributes: Properties of a class.
Methods: Behaviors/functions of a class.

Object-Oriented Principles
1. Encapsulation - Bundles data and methods while protecting data.
2. Abstraction - Hides implementation details.
3. Inheritance - Allows one class to inherit from another.
4. Polymorphism - Same interface with different implementations.

Advantages
Improves code reusability, maintainability, scalability, documentation, and communication among developers.

Conclusion
Class diagrams provide a clear view of system structure, while OOP principles help build robust, maintainable, and reusable software systems.